



Predict the Readiness of your Crew.

THE PROBLEM

Fatigue in aviation is measured too late. Duty-time limits are averages applied to everyone. Fatigue reports depend on a tired person volunteering — and are chronically under-filed. Biomathematical models predict the average pilot on a roster — **they have never met one of your pilots.**

By the time the reactive process finds the fatigue, it isn't a prediction anymore. It's a record.

THE SOLUTION

A short, consent-based readiness check-in before duty. It produces a private **Readiness Score, owned by the crew member alone**, cryptographically sealed at source by Mithra by ShelterZoom.

Operators see aggregate readiness by base, fleet, route and duty pattern — and calibrate predicted fatigue against measured readiness. **Never the person.**

72.9%

European pilots reporting insufficient rest to recover between duties — ECA survey, n≈6,900, 2023

7/10

Saudi commercial pilots reporting burnout — Cureus 2023, n=321, OLBI

56%

UK pilots admitting they have fallen asleep in the cockpit — BALPA, n=500

Predict your readiness — **before reactive methods.**

CONSCIENCEIQ.AI



Predictive Readiness Scoring.

THE FRMS FIT

ICAO's FRMS framework has four components. Conscience IQ is not a fifth — it is a **measured input inside the two that need it most**: a proactive individual readiness signal feeding Fatigue Risk Assessment, and aggregate calibration data feeding FRMS Assurance.

1 • Policy & Documentation

Untouched

2 • Fatigue Risk Assessment

A measured, proactive readiness signal — beside self-reports and surveys ← **CIQ INPUT**

3 • FRMS Assurance

Predicted fatigue vs measured readiness — validate the models you already run ← **CIQ INPUT**

4 • FRMS Promotion

Untouched

THE FLOW

01

CHECK-IN

A human-like AI voice agent guides a short pre-duty conversation — desktop, crew-room tablet, or mobile.

02

ASSESS

Structured, validated fatigue instruments; PVT-informed alertness measures used in aviation research.

03

SENSE

With consent: response latency, attention, voice tone — plus HR & HRV via mobile or wearables.

04

REPORT

A private Readiness Score — load, fatigue, recovery — with cited reasoning the crew member can question.

05

ACT

Aggregate insight for rostering, ops and safety leaders; recovery pathways for the individual.

WHAT IT MEASURES

Cognitive Load

Attention & Latency

Fatigue Load

Recovery State

Stress Response

Burnout Risk

Challenge-Skill Balance

THE CHECK-IN

Five minutes, before duty. The conversation introduces validated fatigue and workload questions while — with consent — ancillary signals add per-question context: how attention, latency and physiological state shift across the assessment. **The difference between what a person reports and what their state is showing.**



Private to the crew. **Aggregate for the operator.**

PRIVACY BY DESIGN

CREW MEMBER · PRIVATE

- A **private readiness report**, cryptographically sealed at source — owned by them alone
- Recovery & recalibration pathways after long-haul, multi-sector and night operations
- **No exposure** of the personal report to the operator — ever

OPERATOR · AGGREGATED

- Readiness by **base, fleet, route or duty pattern** — never the person
- See which pairings degrade recovery; adjust standby, layovers & recovery windows
- **Calibrate biomathematical models** — predicted vs measured

THE TRUST LAYER · SHELTERZOOM / MITHRA

Sealed at Source

Cryptographic hashing, tamper-proof evidence, immutable integrity — Gartner-recognized tokenization, ~80 patents worldwide.

Cited Reasoning

Every recommendation grounded in verified, peer-reviewed sources via a Single Source of Truth — and the crew member can question the report directly.

Simulation Insight

An ancillary human-state layer during full-flight simulator sessions, LOFT scenarios and drills — play-by-play, decision-by-decision.

THE GUARDRAILS

Complements FRMS — never certifies fitness to fly. Go / no-go authority remains with the crew and the operator's regulated FRMS, SMS and aeromedical processes. ICAO's fatigue-management SARPs cover flight crew, cabin crew and air traffic controllers; MRO applications draw on adjacent aviation human-factors research. CIQ is a wellness & readiness tool — never a medical diagnosis, competency judgement, or surveillance.

A 30-day readiness pilot: **one fleet, one base, one duty pattern.** CONSCIENCEIQ.AI